Audiovisual Multimodal Cough Data Analysis for Tuberculosis Detection

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***Abstract*—Early detection of tuberculosis (TB) remains a critical challenge. This research presents a novel multimodal approach utilizing audiovisual information from cough recordings to detect TB. We move beyond traditional image-based methods (such as sputum smear microscopy and chest X-rays) and examine the feasibility of leveraging cough recordings to differentiate TB cases. Two main audio processing techniques, namely Mel-Spectrograms and Mel-Frequency Cepstral Coefficients (MFCCs), are employed for feature encoding of audio recording in deep learning models for TB classification. Our proposed methods leverage a large challenge dataset comprising clinical data from over 1,105 participants and more than 502,252 cough recordings. Notably, a simple 1D Convolutional Neural Network (CNN) trained on MFCC features achieves an accuracy of 91%, surpassing the World Health Organization's (WHO) requirements for TB screening tests. Our findings underscore the potential of MFCC features and 1D CNNs for accurate TB detection utilizing cough sound data. This approach adheres to the Occam's Razor principle, which favors simpler models (such as 1D CNNs) when they yield comparable results. This research paves the way for further study in diverse populations and facilitates the development of accessible TB screening solutions, especially in resource-limited settings where only cough recordings are feasible, thereby emphasizing its notable real-world impacts.**

***Index Terms*—AI in health, audiovisual data, CNN models, holistic methods, Mel-Spectrogram, MFCC, sustainable AI, TB**

I. Introduction

Tuberculosis (TB), an infectious disease caused by Mycobacterium tuberculosis, primarily affects the lungs. The pathogen is transmitted through the air when infected individuals cough, sneeze, or expectorate. TB is ranked as the second leading infectious cause of mortality globally, exceeding HIV and AIDS, with an estimated 10.6 million individuals contracting the disease in 2022 alone. This global health crisis affects individuals of all genders and ages worldwide. In 2022, TB resulted in approximately 1.3 million fatalities. While this disease remains a significant public health threat, there's a beacon of hope: TB is both curable and preventable. To effectively combat this disease, experts estimate an annual investment of US$13 billion is needed, encompassing prevention, diagnosis, treatment, and care initiatives [1], [2]. Early detection is essential for mitigating transmission and improving patient outcomes, but traditional methods like sputum tests frequently encounter limitations in sensitivity, speed, and infrastructure requirements [3], [4]. These limitations result in a significant number of undiagnosed cases, hindering effective control efforts [5]. Current research is constrained by small and homogeneous populations, limiting the applicability of AI tools. There is a need for more diverse studies to develop accurate AI algorithms capable of differentiating TB coughs from non-TB coughs across different demographics. This approach has the potential to enhance the efficacy of AI in

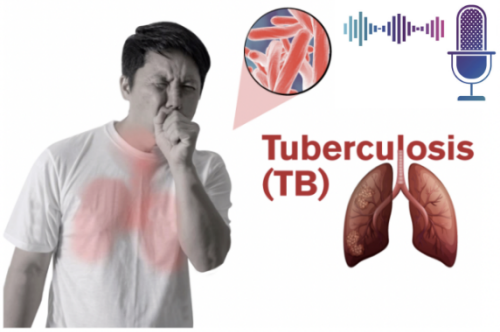

Fig. 1: Individuals with Tuberculosis (TB) Record Coughs Using Mobile Microphones.

addressing TB [6], [7]. The efforts to combat tuberculosis (TB) are significantly advanced by the CODA TB DREAM Challenge. This innovative initiative addresses TB diagnosis by harnessing the power of AI and cough analysis. Participants from seven countries experiencing a persistent cough for two weeks are recruited. Subjects utilize a special app (*Hyfe* Research App) to record their coughs, as shown in Fig. 1, and undergo comprehensive TB evaluations, including lab tests, doctor checkups, and background details [8]. This comprehensive dataset is subsequently made available to the public. AI experts globally are invited to develop algorithms that analyze cough sounds and other information to detect TB. This global collaboration has the potential to significantly enhance TB detection, facilitating faster diagnosis and better patient outcomes. This study examines the extensive CODA TB dataset [8] – over 700,000 coughs from 1,100 participants to analyze cough sounds and investigate the potential of audio information for TB detection. First, the data, including medical details and cough recordings, is meticulously organized. Preliminary analysis has revealed differences in reported symptoms between people with and without TB. For example, symptoms such as weight loss, fever, night sweats, and hemoptysis are observed to be more prevalent in the TB-positive cohort. These findings indicate that integrating cough sound analysis with symptomatic data may enhance the accuracy of TB detection.

This research further investigates the potential of advanced machine learning algorithms for cough sound analysis. We employ two feature extraction techniques (Mel-Spectrograms and MFCCs) to extract meaningful features from cough recordings. These features are subsequently used to develop robust models capable of accurately differentiating TB vs. non-TB, i.e. TB+ / TB- cases. We propose to adapt four deep learning models (1D CNN, 2D CNN, VGG16, and ResNet50) to identify the most effective approach for TB classification. Furthermore, we elucidate factors driving TB prediction and enhance the interpretability of our findings, hence promoting trust and transparency in the diagnostic process.

II. Related Work

The TB disease remains a significant global health concern, with millions of cases reported annually. Existing research [9] investigates the potential of cough analysis using machine learning and deep learning algorithms for automated TB detection. Historically, chest X-ray imaging has been the primary method for TB diagnosis. However, this approach has limitations. Interpreting X-rays requires trained radiologists and subtle abnormalities may be overlooked, as documented in the literature [9]. Furthermore, X-ray imaging exposes patients to ionizing radiation, raising concerns regarding patient safety, especially for repeated testing. Recent research has explored alternative approaches for TB detection that address the limitations of X-ray imaging. Cough analysis emerges as a promising alternative, which offers several advantages. Firstly, it is non-invasive. Cough analysis avoids radiation exposure. Secondly, it facilitates remote monitoring, Cough recordings can be collected without direct patient contact, enabling telemedicine applications. Thirdly, it is a cost-effective approach. Recording and analyzing cough sounds requires minimal equipment compared to X-ray imaging.

Machine learning and deep learning algorithms have demonstrated promising results in cough analysis for various respiratory diseases, including pneumonia and COVID [10], [11], [12]. These algorithms can automatically extract features from cough recordings that are potentially indicative of specific diseases [13], [14]. Many studies have investigated the application of machine learning and data mining techniques for challenging real-world problems, including earlier work by our own research groups [15], [16], [17], [18]. Tsai et al. (2018) employed Mel-frequency Cepstral coefficients (MFCCs) features extracted from cough recordings and achieved an accuracy of 82.2% using a Support Vector Machine (SVM) classifier for TB detection [19]. Cho et al. (2017) utilized convolutional neural networks (CNNs) trained on Mel-Spectrogram representations of cough sounds and reported an accuracy of 87.1% for TB classification [20]. Iwendi et al. (2020) compared various machine learning algorithms, including Random Forests and K-Nearest Neighbors, using MFCC features and achieved an accuracy of 86.3% for TB detection [21]. These studies demonstrate the potential of machine learning and deep learning techniques for automated TB detection using cough analysis. However, there still lacks a comprehensive comparison of deep learning models with various input encoding techniques for improvement in TB detection accuracy and generalizability across diverse populations and cough characteristics.

Our Contribution: This paper is novel in terms of proposing to investigate two feature extraction techniques (Mel-Spectrograms and MFCCs) and evaluating the performance of four deep learning models (1D CNN, 2D CNN, VGG16, and ResNet50) for TB classification using cough recordings in a non-invasive manner. The study underscores the importance of validation and generalizability by assessing the models on external datasets.

III. Data Description and Preprocessing

The CODA TB dataset is collected from health centers across seven continents spanning multiple continents (India, Philippines, South Africa, Uganda, Vietnam, Tanzania, and Madagascar). This international effort recruits participants over 18 years old seeking help at outpatient clinics for a persistent cough lasting at least 2 weeks – a hallmark symptom of TB. Table I presents a statistical overview of a dataset, comprising cough recordings used for Tuberculosis (TB) detection. The table summarizes the data for two groups: people with TB (TB+), and those without (TB-). The CODA TB dataset also incorporates comprehensive clinical data including TB test results, demographics (age, gender, ethnicity), medical history (smoking status, HIV status, prior TB occurrence), and reported symptoms (cough duration, fever, night sweats) (Table II). This comprehensive collection

Features	*TB+*	*TB-*	*Total*
Participants	297	808	1105
Total coughs	443707	280987	724694
Avg. no. of coughs / participant	~1494	~348	~655

TABLE I: Statistical overview of CODA TB dataset

Participants Demographics	TB Negative	TB Positive
Age in Years		
Mean±SD	42.06 ±15.28	37.55 ±14.85
Range	18-85	18-83
Sex		
Male	393(49%)	195(49%)
Female	415(51%)	202(51%)
Anthropometrics		
Height (CM)	160.99±8.79	163.80 ±8.49
Weight (KG)	59.84±14.41	51.84±9.24
BMI (KG/M^2)	23.1	19.3
Heart Rate	82.94±14.27	94.95±19.61
Temperature ©	36.64±0.46	36.96±0.66
Prior Illness		
Prior TB Exposure	151(19%)	48(16%)
P-TB Diagnosis	136(17%)	44(15%)
EP-TB Diagnosis	13(2%)	4(1%)
Presenting Symptoms		
Weight Loss	397(49%)	228(77%)
Fever	298(37%)	199(67%)
Night Sweats	295(37%)	189(62%)
Hemoptysis	84(10%)	64(22%)
Cough Duration / Day (SD)		
Reported at presentation	44.73±56.74	53.29±49.51
Cough Audio(n)		
Solicited Cough	6,842	2,930
Longitudinal Cough	274,145	440,777

TABLE II: Demographic features in cough+metadata experiment

of clinical data enables the investigation of the complex relationships among TB presentation, disease severity, and potential cough variations [1], [22]. AI techniques can effectively capture the learning strategies of scientists in a given domain [23], particularly in health-related contexts [24]. Moreover, the potential for identifying novel biomarkers for TB diagnosis through integrated data analysis underscores the significance of this dataset. However, this valuable resource presents certain challenges. These challenges include missing data points, inconsistencies in reporting, and variations in diagnostic protocols across different healthcare settings. To ensure the quality and reliability of our results, we comprehensively clean and standardize the data, ensuring fitness for robust analysis.

Data Balance - Mitigating Participant Representation Disparities: Although the dataset exhibits a class imbalance favoring TB-positive cases, a more significant challenge is the disproportionate distribution of cough recordings across participants. Some individuals, irrespective of TB status, contribute an anomalously high number of cough recordings (e.g., 71,000 recordings). This biases the training process, as machine learning models tend to prioritize frequently observed patterns. To mitigate this issue, we focus on addressing the imbalance in recordings per participant, rather than the overall class imbalance. This approach balances the dataset for training while preserving valuable participant information [25]. Participants with exceptionally high recording counts are omitted from the analysis, with the maximum number of recordings per participant capped at 990. To maintain a good representation in the training data, at least 990 randomly selected recordings are included from each participant. This approach effectively mitigates bias while retaining valuable individual data. We strategically avoid excluding participants, a consideration particularly crucial for the underrepresented TB+ class. By setting a minimum threshold of 990 recordings per participant, a balanced dataset suitable for machine learning algorithms is achieved, as illustrated in Table III.

Features	*TB+*	*TB-*	*Total*
Participants	297	808	1105
Total coughs	221265	280987	502252
Avg. no. of coughs / participant	~745	~348	~455

TABLE III: Data distribution after removing outliers

Preserving Participant Insights: While outliers are removed, participants are still retained for two major reasons. (1) *Individual Variations:* Cough patterns vary between individuals. Retaining recordings ensures the model is exposed to these variations, potentially aiding in capturing subtle cough characteristics relevant to TB classification. (2) *Participant-Level Context:* The quantity of recordings may inherently contain valuable information. For example, a participant with a substantially higher number of recordings may indicate a more severe cough condition. Retaining a subset of recordings enables the model to potentially learn from this context. This approach balances the dataset for training while preserving valuable information related to individual participants and potential cough-related insights.

IV. Methods

To unlock the hidden secrets within cough recordings, we employ two feature engineering approaches: Mel-Spectrograms and Mel-Frequency Cepstral Coefficients (MFCCs), as illustrated in Fig 2. These techniques transform the raw audio data into visual representations that emphasize the cough's frequency content and characteristics.

Mel-Spectrogram: The first approach employs Mel-Spectrograms, a visual representation of the cough sound's frequency content over time. We explore various deep learning models, including 1D and 2D CNNs, VGG16, and ResNet50, to analyze these Spectrograms. These models demonstrate proficiency in identifying patterns within images, making them well-suited for extracting informative features from the visual cough representations. This process is formulated in Algorithm 1.

MFCC: The second approach utilizes MFCCs, which characterize the spectral envelope of the cough sound.

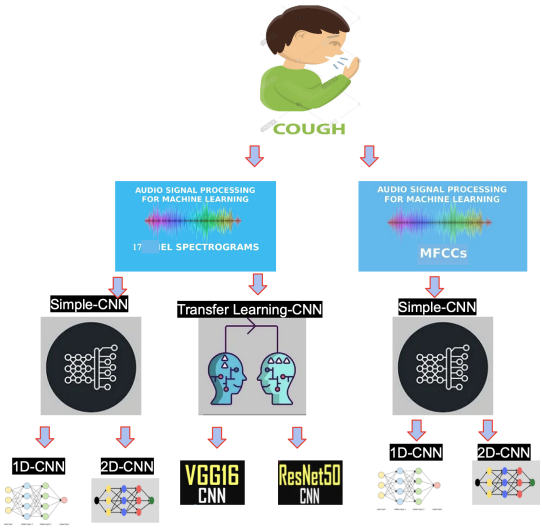

Fig. 2: Framework for Automated Tuberculosis Detection via Cough Analysis

Algorithm 1: Computing Mel-Spectrogram

Data: Audio signal x, Number of Mel filters M, Window size W, Hop length H, Sampling rate F_s
Result: Mel-Spectrogram MS
1: Load audio signal x from the dataset
2: Apply pre-emphasis to the audio signal (optional): $y[n] = x[n] - \alpha \cdot x[n-1]$
3: Divide the pre-emphasized signal into frames of window size W and hop length H
4: **for** *each frame* **do**
  5: Apply a window function (e.g., Hann) to the frame: $windowed_frame = window \cdot frame$
end
6: Compute the Mel-Spectrogram using librosa.feature.melSpectrogram: $MS = \text{librosa.feature.melSpectrogram}(y = x, sr = F_s, n_mels = M, window = window, hop_length = H)$
7: Return the Mel-Spectrogram MS

MFCCs are a well-established technique in audio analysis [26], [27]. Various deep learning models, particularly one-dimensional and two-dimensional CNNs, are adapted to analyze these features. These CNNs excel at processing sequential data like MFCCs, enabling them to discern informative patterns from the cough's spectral characteristics. This process is formulated in Algorithm 2.

Feature Engineering: This process involves transforming the raw audio signal into a format suitable for analysis by deep learning models. This is accomplished by extracting Low-Level Descriptors (LLDs) from each audio frame and then applying statistical operations to compress these features into a more manageable format. This condensed representation allows the deep learning models to focus on the most informative aspects of the cough sound. To enhance the generalizability and robustness of the findings, the pre-

Algorithm 2: Computing MFCCs

Data: Audio signal x, Pre-emphasis coefficient α, Number of Mel filters M, Desired number of MFCC coefficients N
Result: MFCC features $MFCC$
1: Load audio signal x from the dataset
2: Apply pre-emphasis to the audio signal: $y[n] = x[n] - \alpha \cdot x[n-1]$
3: Divide the pre-emphasized signal into frames of length L with hop length H
4: **for** *each frame* **do**
  5: Compute the magnitude spectrum using Fourier Transform: $X[k] = \text{FFT}(y[n])$
  6: Create Mel filter banks using librosa.filters.mel (refer to librosa documentation for arguments)
  7: Apply Mel filters to the magnitude spectrum: $Mel_filtered_spectrum = Mel_filters \cdot X[k]$
  8: Compute MFCC features using librosa.feature.mfcc: $MFCC_coeffs = \text{librosa.feature.mfcc}(y = Mel_filtered_spectrum)$
  9: Keep the first N coefficients of $MFCC_coeffs$ as MFCC features
end
10: Store the MFCC features $MFCC$ for further analysis or classification

Category	***TB+***	***TB-***
Train (75%)	165948	210740
Validation (5%)	11063	14049
Test (20%)	44254	56198
Total	221265	280987

TABLE IV: Data splitting (unit: number of recordings)

processed dataset is systematically partitioned into training (75%), validation (5%), and testing (20%) sets. Critically, we maintain class balance within each set, ensuring the model is trained on a representative distribution of TB+ and TB- cough recordings shown in Table IV.

A. Optimizing Feature Representation: Mel-Spectrogram Conversion Approach

We leverage image classification techniques to differentiate TB+ from TB- cases. However, feeding raw audio signals directly into the model can be computationally expensive. To address this, we propose an approach as shown in Fig. 3 that converts cough recordings into Mel-Spectrograms represented as NumPy arrays. Mel-Spectrograms provide a visually informative representation of the frequency content over time within a cough recording. These spectral representations, analogous to "cough fingerprints", are particularly useful for image classification tasks [28]. By utilizing the Librosa library, we efficiently convert audio signals into Mel-Spectrogram arrays. Librosa achieves this by dividing the audio into the frequency domain and applying Mel filters

to capture the cough's energy distribution across different frequency bands.

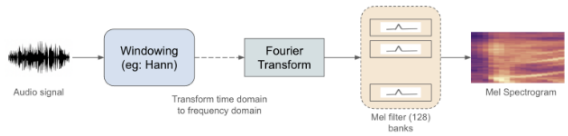

Fig. 3: Proposed Method Approach 1 - Comprehensive Pipeline for TB Detection with Mel-Spectrogram conversion.

This approach offers two main advantages. (1) Reduced Computational Cost: NumPy arrays are optimized for numerical computations, leading to faster model training compared to raw audio data. (2) Efficient Memory Management: NumPy arrays provide efficient memory handling, which is crucial for large datasets.

It is noteworthy that a diverse array of methods has been employed in the literature for large-scale data processing, involving data mining over biological data [29] and image processing over complex data types [30]. Likewise, in this study, we investigate the performance of four commonly used deep learning models to systematically assess the efficacy of Mel-Spectrogram features for TB+ vs. TB- classification (Approach 1). We now introduce each model's architectures, training processes, and key findings.

One-Dimensional Convolutional Neural Network (1D CNN) with Mel-Spectrograms: This model (Fig. 4) serves as a baseline, utilizing a sequential architecture with two hidden convolutional layers. It leverages ReLU activation for efficient learning in hidden layers and Softmax activation in the output layer for multi-class classification. Max pooling is implemented to facilitate downsampling after each convolutional layer, thereby reducing feature dimensionality.

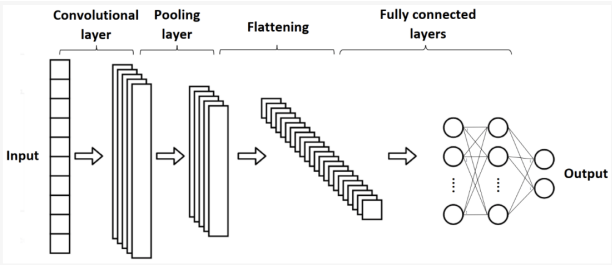

Fig. 4: 1D CNN with Mel-Spectrograms [32]

Two-Dimensional Convolutional Neural Network (2D CNN) with Mel-Spectrograms: Building upon the 1D CNN, we implement a more complex 2D CNN architecture with three hidden convolutional layers. This model (Fig. 5) also utilizes ReLU activation in hidden layers, but employs a Sigmoid activation function in the output layer, optimized for binary classification tasks. Average pooling is used for downsampling after each convolutional layer. The training process exhibits a smooth convergence pattern, with a significant initial drop in loss and a corresponding increase in accuracy, reaching a stable state around the 20th epoch. Early stopping is implemented to prevent overfitting, ensuring the model generalizes well to unseen data.

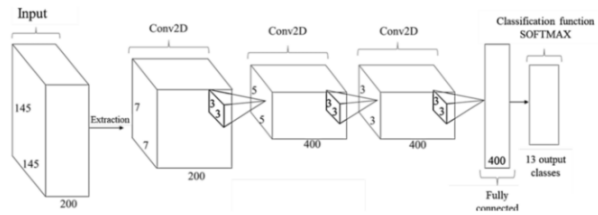

Fig. 5: 2D CNN with Mel-Spectrograms [31].

Transfer Learning with VGG16: To harness the power of pre-trained models, we leverage transfer learning with VGG16, a deep convolutional neural network pre-trained on the massive ImageNet dataset. VGG16 excels at extracting informative features from images. In this approach (Fig. 6), we freeze the top layer of the pre-trained VGG16, preserving its learned features. Subsequently, a series of fully connected dense layers is added on top of the pre-trained network. These new layers are trained using our Mel-Spectrogram data to fine-tune VGG16 for TB classification. This approach leverages VGG16's feature extraction capabilities while adapting it to the specific task of TB detection.

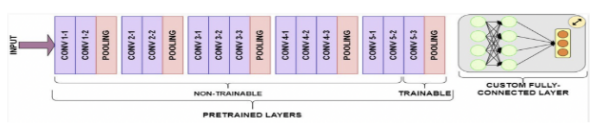

Fig. 6: VGG16 architecture with Mel-Spectrograms [32].

Transfer Learning with ResNet50: We further investigate transfer learning via ResNet50 (Fig. 7), a more complex pre-trained convolutional neural network on ImageNet. Similar to VGG16, features extracted from ResNet50 are passed through a dense layer to generate the final TB classification prediction. ResNet50's architecture potentially provides richer feature representations compared to VGG16, which could lead to improved TB detection accuracy. The performance of VGG16 and ResNet50 for TB classification with Mel-Spectrograms is systematically evaluated and compared.

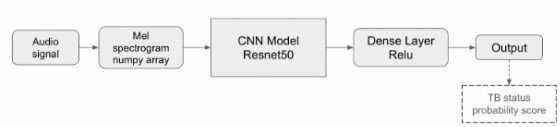

Fig. 7: ResNet50 architecture with Mel-Spectrograms [33].

B. Optimizing Feature Representation: Mel-Frequency Cepstral Coefficients (MFCC) Extraction

As an alternative approach, we propose to investigate Mel-Frequency Cepstral Coefficients (MFCCs), as shown in Fig. 8. Unlike Mel-Spectrograms, MFCCs directly capture the perceptually relevant spectral shape of the audio signal, focusing on frequencies crucial to human hearing. This compressed representation offers two key advantages. (1) Enhanced Computational Efficiency: The extraction of MFCCs is computationally more efficient compared to the generation of Mel-Spectrograms, rendering it appropriate for real-time or resource-constrained environments. (2) Compact Feature Representation: MFCCs provide a more concise feature set compared to Spectrograms, potentially leading to improved model training efficiency.

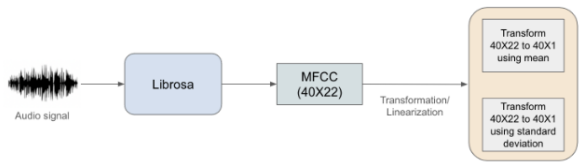

Fig. 8: Proposed Method Approach 2 - Comprehensive Pipeline for TB Detection using Mel-frequency Cepstral coefficients (MFCCs).

1D CNN with MFCC: We propose to employ a model with a sequential architecture of two hidden convolutional layers specifically designed to process 1D feature vectors representing the MFCCs. It is depicted in Fig. 9. The key components of this model include the following. (1) Sequential Architecture: Layers are stacked sequentially, with the output of one layer serving as input to the subsequent layer. (2) Downsampling: Max pooling, implemented after each convolutional layer, is a widely adopted technique for reducing feature map dimensionality while preserving salient features. (3) Output Layer: The final dense layer with a Softmax activation function is suitable for multi-class classification (TB+ vs. TB-). The Softmax function generates class probabilities, indicating the likelihood of each class for a given cough sample.

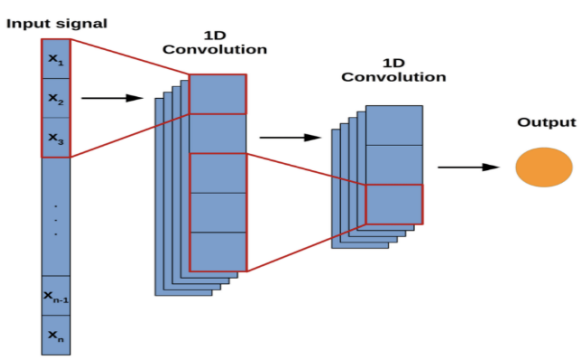

Fig. 9: 1D CNN with MFCC [33].

2D CNN with MFCC: We further propose to explore a 2D CNN architecture with MFCC features, as shown in Fig. 10. This model utilizes Librosa to convert audio recordings into MFCCs, resulting in a 3D tensor representation (time, frequency, coefficients). The key components of this model are detailed as follows. (1) Architecture: Sequential CNN with three hidden convolutional layers. (2) Hidden Layers: Similar to the 1D CNN, each hidden layer utilizes the ReLU activation function. (3) Downsampling: Average pooling is used after each convolutional layer for dimensionality reduction. (4) Output Layer: The final dense layer employs the Sigmoid activation function, suitable for binary classification, i.e. TB vs. Non-TB outputs.

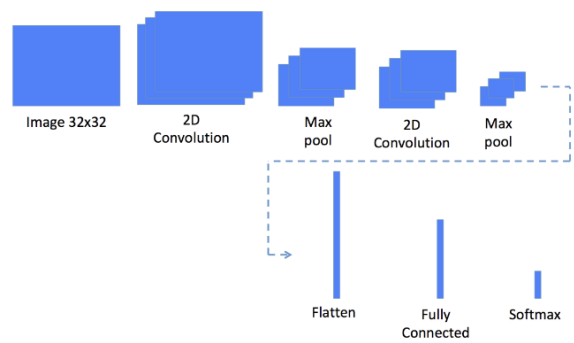

Fig. 10: 2D CNN with MFCC [31].

Enhancing Model Generalizability through Cross-Validation: To optimize the models' generalizability and mitigate overfitting, we employ k-fold cross-validation *(k = 5)*. The dataset is partitioned into *k* folds. For each fold, the model is trained on *(k-1)* folds and evaluated on the remaining unseen validation fold, utilizing metrics such as accuracy, precision, recall, and F1-score. This training-evaluation cycle was repeated for all *k* folds. The performance metrics from each round were averaged to yield a more robust estimate of model generalizability on unseen data.

V. Results & Discussion

A. Performance of Models

This section presents the performance analysis of models utilizing Mel-Spectrogram features. The following observations are noted. Table V summarizes the performance comparison of various deep learning models for TB classification using Mel-Spectrograms.

Notably, 1D CNN with MFCCs model demonstrates superior overall performance, outperforming models employing Mel-Spectrograms. This suggests that processing the raw audio signal and directly learning features might be a more effective approach for TB classification in this dataset. In fact, the 2D CNN with MFCCs model underperforms the 1D CNN model, possibly due to the increased complexity of the 2D representation being sub-optimal for directly capturing relevant patterns from the audio signal. Table VI summarizes the performance comparison of various deep learning models for TB classification using MFCCs.

Model	Precision (TB)	Recall (TB)	Precision (Non-TB)	Recall (Non-TB)	F1 Score (TB)	Accuracy	AUC
Simple 2D CNN	0.58 ± 0.04	0.23 ± 0.02	0.74 ± 0.03	0.93 ± 0.01	0.33 ± 0.03	0.72 ± 0.02	70% ± 3%
Simple 1D CNN	**0.71 ± 0.02**	**0.62 ± 0.04**	**0.85 ± 0.01**	**0.89 ± 0.02**	**0.66 ± 0.02**	**0.81 ± 0.01**	**~81% ± 1%**
VGG16 (Transfer Learning)	0.72 ± 0.03	0.47 ± 0.05	0.8 ± 0.02	0.92 ± 0.01	0.57 ± 0.04	0.79 ± 0.02	~78% ± 2%
ResNet50 (Transfer Learning)	0.6 ± 0.01	0.55 ± 0.05	0.82 ± 0.02	0.85 ± 0.03	0.58 ± 0.01	0.76 ± 0.01	~79% ± 1%

TABLE V: Performance comparison of Mel-Spectrogram models for TB classification

Model	Precision (TB)	Recall (TB)	F1 Score (TB)	Precision (Non-TB)	Recall (Non-TB)	F1 Score (Non-TB)	Accuracy	AUC
Simple 1D CNN	**0.9 ± 0.02**	**0.84 ± 0.03**	**0.87 ± 0.02**	**0.91 ± 0.01**	**0.95 ± 0.02**	**0.93 ± 0.01**	**0.91 ± 0.01**	**~91% ± 0.5%**
2D CNN	0.66 ± 0.03	0.5 ± 0.04	0.57 ± 0.03	0.81 ± 0.02	0.89 ± 0.03	0.85 ± 0.02	0.77 ± 0.02	~82% ± 0.7%

TABLE VI: Performance comparison of MFCC models for TB classification

B. Discussion on Performance

Key Observations and Alignment with Occam's Razor: Analysis of all models reveals that a simple 1D CNN designed for processing 1D features yielded the best results for TB classification using both Mel-Spectrograms and raw audio signals. This observation aligns with the principle of Occam's Razor, favoring simpler models when they achieve comparable or better performance [34]. Such solutions in favor of simplicity have frequently been adopted in the literature, including in previous studies [35], [36], [37]. Transfer learning approaches might require further fine-tuning or utilizing pre-trained models specifically designed for audio tasks to improve their effectiveness in TB classification. Our findings demonstrate the potential of multiple deep learning models trained on cough sound features with adequate feature extraction to accurately classify TB cases.

Open Issues Emerging from the Study: While this research demonstrates promise, several open issues and opportunities for further studies remain. Examination of the impact of longer cough recordings on classification accuracy is crucial. External validation on broader datasets encompassing diverse populations and TB prevalence rates is essential for generalizability. Incorporating explainable artificial intelligence (XAI) techniques may augment model interpretability, potentially facilitating the discovery of novel audio biomarkers for TB detection. Investigation of one-dimensional audio-specific architectures may potentially enhance performance and mitigate complexity relative to more intricate models or Mel-Spectrogram-based approaches. Further research and development are necessary to refine the models, optimize performance, and ensure generalizability across diverse populations and cough characteristics. Addressing various challenges, e.g. background noise, cough variations, and developing user-friendly recording devices, is necessary for clinical implementation.

AI-Based Medical Analysis: By addressing these open issues and pursuing further studies, we can refine the models for TB classification using cough sounds. This research trajectory possesses the potential to yield a robust, interpretable, and generalizable approach for TB detection, ultimately contributing to advancements in public health [38], [39]. This study presents a significant step towards utilizing AI-powered cough analysis as a valuable tool in the global fight against tuberculosis, thereby making significant impacts on healthcare.

VI. Conclusions and Roadmap

This study examines the potential of deep learning-based classification using cough sound analysis for Tuberculosis (TB) detection. Two feature extraction approaches are investigated: Mel-Spectrograms and MFCC. We evaluate the effectiveness of four neural network models for TB classification using these features.

Our findings reveal the potential of integrating audio data (cough sounds) to improve TB detection accuracy. This research trajectory facilitates the exploration of non-invasive and potentially cost-effective screening tools. We demonstrate the effectiveness of a simple 1D CNN model for TB classification using MFCC features [40]. This finding suggests that directly learning features from the raw audio signal might be more efficient compared to complex architectures or Mel-Spectrogram representations for this specific task. We observe the limited benefits of utilizing transfer learning approaches with pre-trained models such as VGG16 and ResNet50 for TB detection.

Our findings pave the way for many promising avenues. Primarily, there is a need to examine the impact of longer recordings and validate the model on diverse populations. Such an approach could significantly enhance the model's predictive capabilities. Furthermore, the incorporation of uncertainty quantification is crucial to enhance confidence in the model's predictions, thereby facilitating evidence-based clinical decisions. To broaden the scope of this research, the algorithmic scope should be extended, encompassing additional classifiers and conducting comprehensive evaluations to ensure robustness and reliability.

By addressing these future directions, we can refine deep learning models for TB classification using cough sounds. This holds promise for developing a robust, interpretable, and generalizable AI-infused approach to TB detection, ultimately improving public health.

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